SQL-fundamentals:-

Q.1

Employees

id	First_name	last_name	department	salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburs
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Motoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Sol^n

```
SELECT *
from employees;
```

id	First_name	Last_name	department	salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	~~Zulu~~ Motoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q.2 Find all unqui departments

```
SELECT        DISTINCT department
FROM Employees;
```

department
IT
HR
Finance
marketing

Q.3 First and Last names by descending

```
SELECT First_name,
       Last_name
FROM Employees
ORDER BY ~~employee~~ First_name, Last_name DESC;
```

Q.3 Retrieve first_name and last_name ordered by descending

```
SELECT first_name,
       last_name
FROM Employees
ORDERBY salary DESC;
```

First name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q.4 Top 3 highest paid.

```
SELECT id,
       first_name,
       last_name,
       salary
FROM Employees
ORDERBY salary DESC
LIMIT 3;
```

id	first_name	last_name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Q.5 Find employees in IT

```
SELECT id,
       first_name,
       last_name,
       department
FROM Employees
WHERE department = 'IT';
```

id	first_name	Last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q.6 Find employees in Finance with salary >60000

```
SELECT id,
       First_name,
       Last_name,
       department,
       Salary
FROM Employees
WHERE department = 'finance' AND Salary > 60000;
```

Id	FirstName	Last_name	department	Salary
3	Cathy	Zulu	Finance	65000

Q.7 Find employees in HR or Marketing.

```
SELECT id,
       First_name,
       Last_name,
       department
FROM Employees
WHERE department = 'HR' or 'marketing';
```

Id	First_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q.8 Find employees NOT in IT

```
SELECT id,
       first_name,
       last_name,
       department
FROM Employees
WHERE department NOT 'IT';
```

id	First_name	last_name	~~department~~
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q.9 Find employees in IT, HR or Finance

```
SELECT id,
       first_name,
       last_name,
       department
FROM Employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
	Brian	Lee	HR
	Cathy	Zulu	Finance

Q.10 Find Employees in IT with Salary > 65 000 and in the City of Johannesburg.

```
SELECT id,
       first_name,
       last_name,
       department,
       salary,
       city
FROM Employees
WHERE department = 'IT' ~~AND salary > 65000 AND~~
  AND Salary > 65000
  AND City = 'Johannesburg';
```

id	First_name	last_name	department	salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg